

Rafael Alain Rolli

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PROFILE

Biotechnologist and founder with an M.Sc. in Bioengineering from EPFL and hands-on R&D leadership across multiple ventures in Switzerland and Germany. I have run wet labs and the surrounding companies, and I currently lead the science behind an autonomous AI agent for peptide drug discovery, designing experimental pipelines, structured data systems, and AI agents that accelerate research. Having worked as a scientist inside R&D platforms during my day-to-day, I understand what makes them succeed or fail on the bench. I move fluently between scientists, engineers, and executives, and I am at my best turning ambiguous scientific problems into structured, repeatable workflows.

EXPERIENCE

PeptAI | Remote 12/2025 – Present
Science Lead & Co-Founder

PeptAI is a spin-out from the Molecule AG ecosystem.

- Co-founded and led the science behind an autonomous, AI-driven peptide drug-discovery platform. Discovery strategy from target selection through computational design to wet-lab validation.
- Designed the end-to-end discovery pipeline as a structured, staged workflow with defined decision gates and data standards, ensuring reproducibility across computational predictions and experimental results.
- Built AI agents directly into research execution, automating steps that traditionally require manual scientist effort and compressing discovery cycles.
- Translate scientific requirements into vendor-ready specifications for external CRO partners (peptide synthesis, binding and functional assays) and integrate returned data back into the pipeline.
- Work across scientists, engineers, and commercial stakeholders to turn an ambiguous research problem into a repeatable operational process.

Molecule AG | Remote 02/2025 – Present
Head of Science & AI

- Run customer-discovery interviews for BIOS, an AI research tool, translating user needs into prioritized feature requests with the product team.
- Lead biotech partnerships and deal flow, ensuring successful collaboration on research projects.
- Facilitate technology transfer with Tech Transfer Offices and universities, strengthening innovation pathways.
- Source research projects across longevity, brain health, synthetic biology, and peptide drug discovery.
- Foster connections with leading scientists to build collaborative research opportunities.

Infinit Biosystems GmbH | Berlin, Germany 10/2021 – 02/2024
Co-Founder & Head of R&D

- Co-founded and ran an enzymatic biosynthesis company, leading both the wet lab and company operations from formation through multiple funding rounds.
- Designed and executed R&D directly at the bench: enzymatic platform development for psychedelic API biosynthesis, managing the team running experiments.
- Used Benchling as the team's R&D system of record for experiment design and structured data capture.
- Directed a cross-functional team and ensured compliance with European regulatory standards.

Antler | Berlin, Germany 10/2021 – 03/2022
Entrepreneur in Residence

- Developed and pitched scalable biotech solutions, securing pre-seed funding and validating market potential.

Eir Drop Biosciences | Lausanne, Switzerland 07/2020 – 09/2021

Co-Founder

- Co-founded an EPFL spin-off developing scalable cell-free biosynthesis systems for natural ingredients.
- Applied machine-learning models to simulate and validate genetic sequences, accelerating research cycles.

Alponics SA | Martigny, Switzerland 06/2019 – 06/2020

Cannabis Chemist & Laboratory Director

- Managed daily laboratory operations in full compliance with ISO and GMP processes.
- Performed QA/QC analytical testing for product quality, consistency, and efficacy.
- Developed and refined extraction and infusion techniques, including supercritical CO2 extraction.
- Built expertise in the purification, separation, and isolation of compounds.

EDUCATION

École Polytechnique Fédérale de Lausanne (EPFL) 07/2016 – 10/2018

Master of Science (M.Sc.), Bioengineering & Biomedical Engineering

Master thesis: Cardiac Mechanics — University of California, San Diego.

École Polytechnique Fédérale de Lausanne (EPFL) 09/2013 – 06/2016

Bachelor of Science (B.Sc.), Bioengineering & Biomedical Engineering

SKILLS

Languages: Spanish (native), English (fluent), French (fluent), German (professional working proficiency)

Project & Implementation: Project management, requirements gathering, customer & stakeholder management, cross-functional team leadership

R&D & Scientific: Wet-lab R&D, experimental design, GMP / ISO compliance, QA/QC analytical testing, enzymatic biosynthesis

AI & R&D Platforms: AI-driven drug discovery, AI agent development, AI-assisted sequence design, Benchling, BIOS

Leadership & Operations: Strategic planning, operations, fundraising

ACHIEVEMENTS

- Secured funding: raised \$170,000 in pre-seed funding (Antler) and \$250,000 in seed funding (Berlin Founders Fund, health tech).
- Co-invented a cell-free biosynthesis platform leveraging machine learning to optimize small-molecule production for cannabinoids and tryptamines.
- Consulted for biotech startups integrating AI into protein engineering cycles, including sequence design, protocol planning, and autonomous experimentation.
- Led cross-functional teams and partnerships across English, French, Spanish, and German-speaking regions.

CERTIFICATES

Innosuisse Biotech Business Creation — EPFL 01/2020

Intensive program preparing scientists to found and run companies in Switzerland and the EU.

INTERESTS

Building AI agents for everyday use · Handstand training and gymnastics · Hermetic science and alchemy · Blockchain and decentralized-science writing · Community building · Gardening and greenhouse cultivation